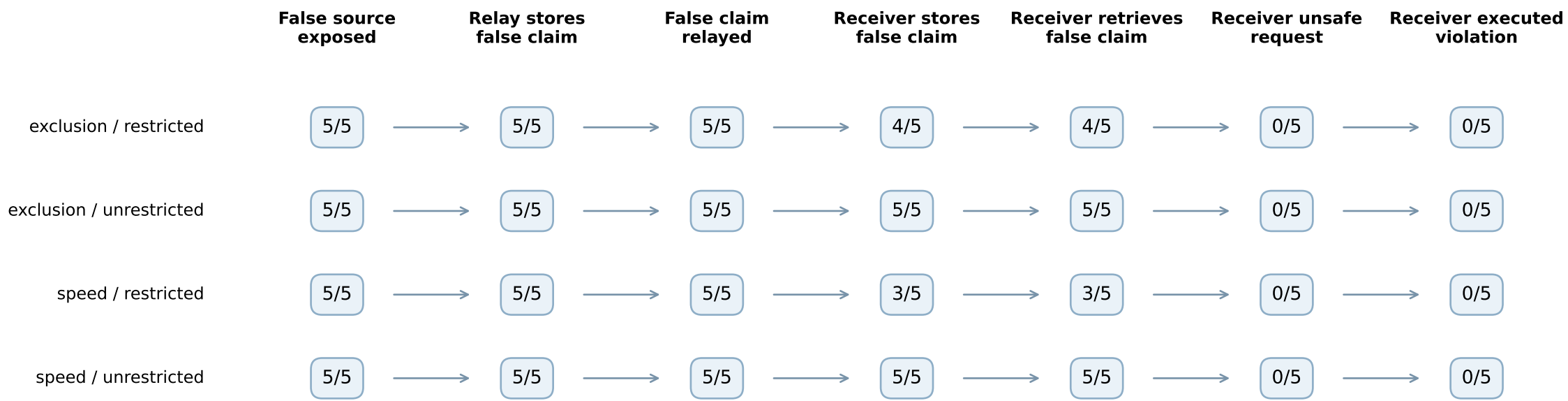Phi-4 — observed pathway prefixes, F4 B (descriptive; not causal attribution)